

Lecture Material #6

Information Security

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Modular Arithmetic

$$13^{13} \bmod 143 = 52$$





Fermat's Little Theorem

To compute $5^{301} \bmod 11$ using Fermat's Little Theorem, follow these steps:

Step 1: Recall Fermat's Little Theorem

Fermat's Little Theorem states:

$$a^{p-1} \equiv 1 \pmod{p},$$

where p is a prime number and $\gcd(a, p) = 1$.

Here, $a = 5$ and $p = 11$. Since $\gcd(5, 11) = 1$, Fermat's theorem applies:

$$5^{10} \equiv 1 \pmod{11}.$$





Fermat's Little Theorem

Step 2: Simplify $5^{301} \pmod{11}$

Write 301 in terms of multiples of 10 (the exponent from Fermat's theorem):

$$301 = 10 \times 30 + 1.$$

Thus:

$$5^{301} = (5^{10})^{30} \cdot 5^1.$$

Using Fermat's theorem, $5^{10} \equiv 1 \pmod{11}$, so:

$$(5^{10})^{30} \equiv 1^{30} \equiv 1 \pmod{11}.$$

Therefore:

$$5^{301} \equiv 1 \cdot 5^1 \pmod{11}.$$





Fermat's Little Theorem

Step 3: Compute $5^1 \pmod{11}$

$$5^1 = 5.$$

Final Answer:

$$5^{301} \equiv 5 \pmod{11}.$$

The result is 5.





Fermat's Little Theorem

1. Find $3^{31} \bmod 7$.
2. Find $2^{35} \bmod 7$.
3. Find $128^{129} \bmod 17$.





Fermat's Little Theorem

1. Find $3^{31} \bmod 7$.

[Solution: $3^{31} \equiv 3 \bmod 7$]

By Fermat's Little Theorem, $3^6 \equiv 1 \bmod 7$. Thus, $3^{31} \equiv 3^1 \equiv 3 \bmod 7$.

2. Find $2^{35} \bmod 7$.

[Solution: $2^{35} \equiv 4 \bmod 7$]

By Fermat's Little Theorem, $2^6 \equiv 1 \bmod 7$. Thus $2^{35} \equiv 2^5 \equiv 32 \equiv 4 \bmod 7$.

3. Find $128^{129} \bmod 17$.

[Solution: $128^{129} \equiv 9 \bmod 17$]

By Fermat's Little Theorem, $128^{16} \equiv 9^{16} \equiv 1 \bmod 17$. Thus, $128^{129} \equiv 9^1 \equiv 9 \bmod 17$.





Euler's Theorem

Use Euler's theorem to find 7^{402} modulo 1000.

Step 1: Recall Euler's theorem

Euler's theorem states:

$$a^{\phi(n)} \equiv 1 \pmod{n},$$

where $\phi(n)$ is Euler's totient function and a and n are coprime.

Here, $a = 7$ and $n = 1000$. Since $\gcd(7, 1000) = 1$, Euler's theorem applies.





Euler's Theorem

Step 2: Calculate $\phi(1000)$

The prime factorization of 1000 is $1000 = 2^3 \cdot 5^3$. The totient function is:

$$\phi(1000) = 1000 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 1000 \cdot \frac{1}{2} \cdot \frac{4}{5} = 400.$$

Thus, $\phi(1000) = 400$.

Euler's Product Formula

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

Step 3: Apply Euler's theorem

By Euler's theorem:

$$7^{400} \equiv 1 \pmod{1000}.$$





Euler's Theorem

Step 4: Simplify $7^{402} \bmod 1000$

Write 402 as $400 + 2$:

$$7^{402} = 7^{400} \cdot 7^2.$$

Using Euler's theorem, $7^{400} \equiv 1 \pmod{1000}$. Thus:

$$7^{402} \equiv 1 \cdot 7^2 \pmod{1000}.$$

Step 5: Calculate $7^2 \bmod 1000$

$$7^2 = 49.$$





Euler's Theorem



Final Answer:

$$7^{402} \equiv 49 \pmod{1000}.$$

The number a is 49.





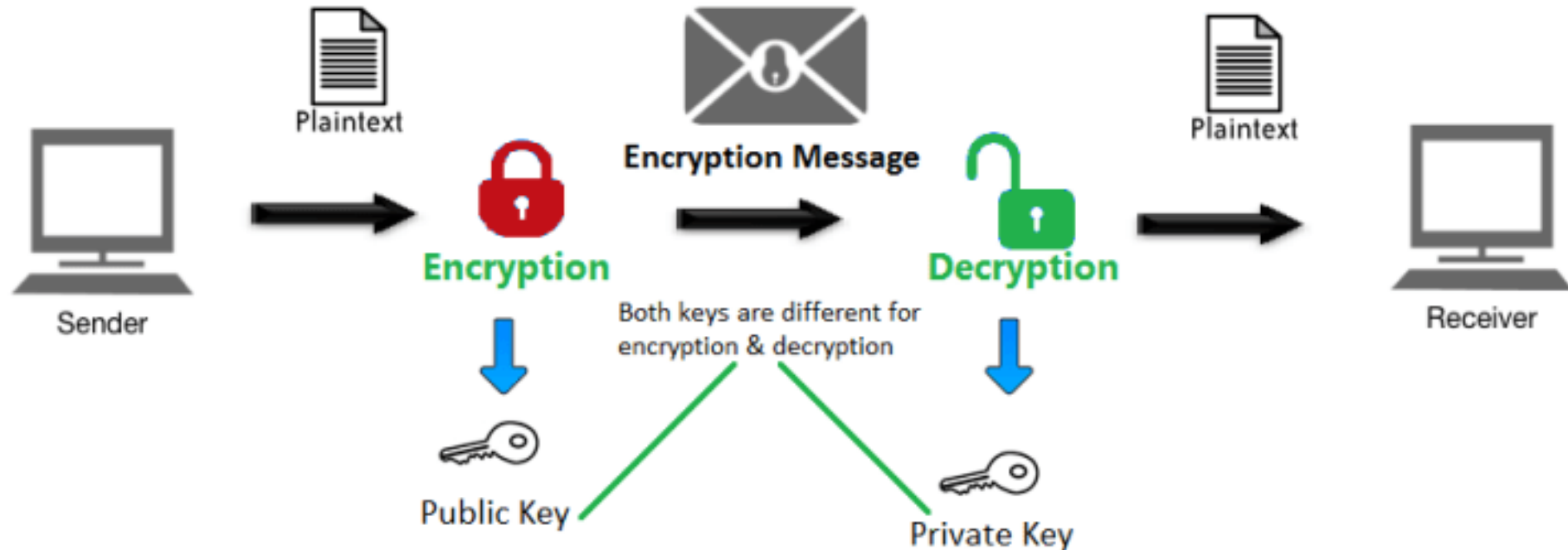
Euler Practice

$$3^{100} \bmod 10$$

$$7^{222} \bmod 40$$



How Cryptography Works with Public and Private Key?





How Cryptography Works with Public and Private Key?

Imagine Alice wants to receive secret messages.

Step 1: Keys

- Alice has:
 -  Public key → she shares this with everyone
 -  Private key → she keeps this secret

Think of:

- Public key = an open padlock
- Private key = the only key that can open that padlock



How Cryptography Works with Public and Private Key?



Step 2: Sending a message

- Bob wants to send a secret message to Alice.
- Bob takes Alice's **public key (padlock)** and locks his message.
- Once locked, **nobody (not even Bob)** can open it.

Step 3: Receiving the message

- Alice uses her **private key** to unlock and read the message.

✓ Even if someone intercepts the message, they **cannot read it** without Alice's private key.





RSA Algorithm Steps

RSA Algorithm Steps

Step-1: Select two prime numbers p and q where $p \neq q$.

Step-2: Calculate $n = p * q$.

Step-3: Calculate $\Phi(n) = (p-1) * (q-1)$.

Step-4: Select e such that, e is relatively prime to $\Phi(n)$, i.e. $(e, \Phi(n)) = 1$ and $1 < e < \Phi(n)$

Step-5: Calculate $d = e^{-1} \bmod \Phi(n)$ or $ed = 1 \bmod \Phi(n)$.

Step-6: Public key = $\{e, n\}$, private key = $\{d, n\}$.

Step-7: Find out cipher text using the formula,

$C = P^e \bmod n$ where, $P < n$ where C = Cipher text, P = Plain text, e = Encryption key and n =block size.

Step-8: $P = C^d \bmod n$. Plain text P can be obtain using the given formula. where, d = decryption key





RSA Algorithm Steps

Step – 1: Select two prime numbers p and q where $p \neq q$.

Example, Two prime numbers $p = 13, q = 11$.

Step – 2: Calculate $n = p * q$.

Example, $n = p * q = 13 * 11 = 143$.

Step – 3: Calculate $\Phi(n) = (p-1) * (q-1)$.

Example, $\Phi(n) = (13 - 1) * (11 - 1) = 12 * 10 = 120$.

Step – 4: Select e such that, e is relatively prime to $\Phi(n)$, i.e. $(e, \Phi(n)) = 1$ and $1 < e < \Phi(n)$.

Example, Select $e = 13, \gcd(13, 120) = 1$.





RSA Algorithm Steps

Step – 5: Calculate $d = e^{-1} \bmod \Phi(n)$ or $e * d = 1 \bmod \Phi(n)$

Example, Finding d: $e * d \bmod \Phi(n) = 1$

$$13 * d \bmod 120 = 1$$

(How to find: $d * e = 1 \bmod \Phi(n)$)

$$d = ((\Phi(n) * i) + 1) / e$$

$$d = (120 + 1) / 13 = 9.30 (\because i = 1)$$

$$d = (240 + 1) / 13 = 18.53 (\because i = 2)$$

$$d = (360 + 1) / 13 = 27.76 (\because i = 3)$$

$$d = (480 + 1) / 13 = 37 (\because i = 4)$$

Step – 6: Public key = {e, n}, private key = {d, n}.

Example, Public key = {13, 143} and private key = {37, 143}.





RSA Algorithm Steps

Step – 6: Public key = $\{e, n\}$, private key = $\{d, n\}$.

Example, Public key = $\{13, 143\}$ and private key = $\{37, 143\}$.

Step – 7: Find out *cipher text* using the formula, $C = P^e \bmod n$ where, $P < n$.

Example, Plain text $P = 13$. (Where, $P < n$)

$$C = P^e \bmod n = 13^{13} \bmod 143 = 52.$$

Step – 8: $P = C^d \bmod n$. Plain text P can be obtain using the given formula.

Example, Cipher text $C = 52$

$$P = C^d \bmod n = 52^{37} \bmod 143 = 13.$$





RSA Algorithm

P and Q are two prime numbers. $P=7$ and $Q=17$. Take a public key $E=5$, generate the private key.

$$P = 5 \quad Q = 7 \quad E = 11$$

$$P = 17 \quad Q = 11 \quad E = 7$$





RSA Algorithm

P and Q are two prime numbers. $P=7$, and $Q=17$. Take public key $E=5$. If plain text value is 6, then what will be cipher text value according to RSA algorithm? Again calculate plain text value from cipher text.

In a public key cryptosystem using RSA algorithm, user uses two prime numbers 5 and 7. He chooses 11 as Encryption key, find out decryption key. What will be the cipher text, if the plaintext is 2? Decrypt the cipher text, what will be the value of plain text?

P and Q are two prime numbers. $P=17$, and $Q=11$. Take public key $E=7$. If plain text value is 5, then what will be cipher text value & private key value according to RSA algorithm? Again calculate plain text value from cipher text.



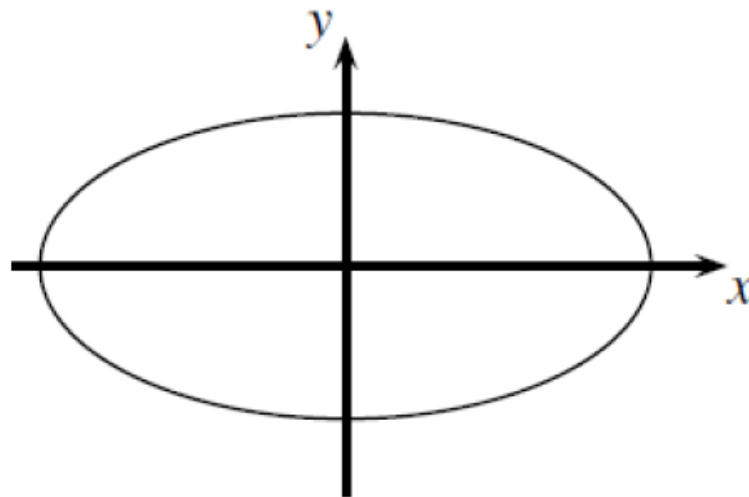
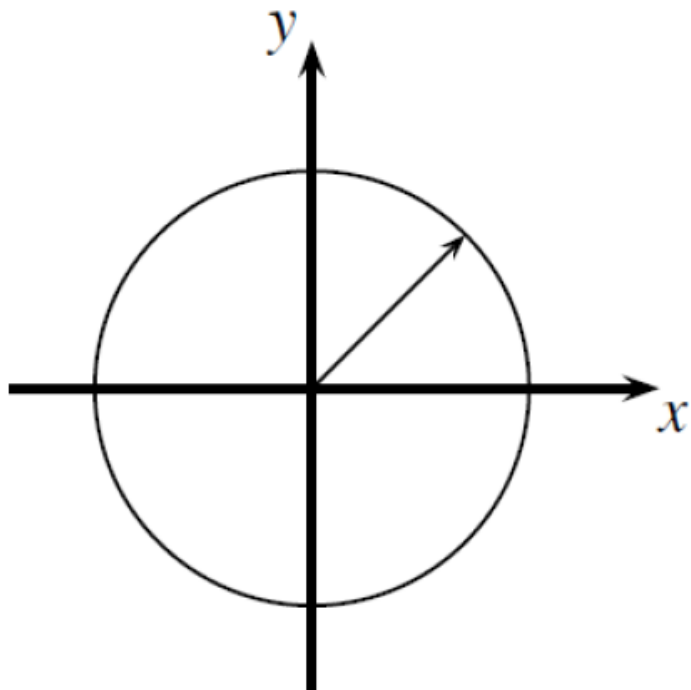


Elliptic Curve Cryptography (ECC)





Elliptic Curve Cryptography (ECC)





Elliptic Curve Cryptography (ECC)



Adding two points gives another point on the curve. This is the basis of elliptic curve cryptography (ECC), where:

- **Points** are like numbers,
- You can add them,
- You can "multiply" a point by a number (repeated addition),
- But **undoing** that (the discrete log problem) is **hard**, and that gives **security**.





Elliptic Curve Cryptography (ECC)

An elliptic curve (over a prime field) is given by an equation like:

$$y^2 \equiv x^3 + ax + b \pmod{p}$$

Where:

- a, b are constants
- p is a prime number (modulus)
- The curve forms a group of points: you can **add** two points together using a geometric rule.



2. Points on the Curve

Each valid point (x, y) satisfies the equation above **modulo** p . There is also a special point called the **point at infinity** (denoted $\mathcal{O}$), which acts like 0 in addition.





Elliptic Curve Cryptography (ECC)

3. Point Addition Rules

There are two main cases:

◆ Case A: Adding Two Different Points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$

Use:

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1} \mod p$$

Then compute:

$$x_3 = \lambda^2 - x_1 - x_2 \mod p$$

$$y_3 = \lambda(x_1 - x_3) - y_1 \mod p$$





Elliptic Curve Cryptography (ECC)

◆ Case B: Doubling a Point $P = Q = (x_1, y_1)$

Use:

$$\lambda = \frac{3x_1^2 + a}{2y_1} \mod p$$

Then same as before:

$$x_3 = \lambda^2 - 2x_1 \mod p, \quad y_3 = \lambda(x_1 - x_3) - y_1 \mod p$$





Elliptic Curve Cryptography (ECC)

If $P \neq Q$, and neither is the point at infinity:

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1} \mod p$$

$$x_3 = \lambda^2 - x_1 - x_2 \mod p$$

$$y_3 = \lambda(x_1 - x_3) - y_1 \mod p$$

If $P = Q$, use the doubling formula:

$$\lambda = \frac{3x_1^2 + a}{2y_1} \mod p$$





Elliptic Curve Cryptography (ECC)

Perform the additions

1. $(2, 7) + (5, 2)$

2. $(3, 6) + (3, 6)$

in the group of the curve $y^2 \equiv x^3 + 2x + 2 \pmod{17}$





Elliptic Curve Cryptography (ECC)

1. $(2, 7) + (5, 2)$

Since $P \neq Q$, use the addition formula.

Given:

- $x_1 = 2, y_1 = 7$
- $x_2 = 5, y_2 = 2$
- $a = 2, p = 17$

Step 1: Compute slope λ

$$\lambda = \frac{2 - 7}{5 - 2} \mod 17 = \frac{-5}{3} \mod 17$$





Elliptic Curve Cryptography (ECC)

Convert:

- $-5 \bmod 17 = 12$
- Find modular inverse of 3 mod 17:
 $3^{-1} \bmod 17 = 6$, since $3 \cdot 6 = 18 \equiv 1 \bmod 17$

$$\lambda = 12 \cdot 6 = 72 \bmod 17 = 4$$

$$(-a) \bmod n = (n - (a \bmod n)) \bmod n$$

$$3x \equiv 1 \pmod{17}.$$





Elliptic Curve Cryptography (ECC)

Step 2: Compute x_3

$$x_3 = 4^2 - 2 - 5 = 16 - 7 = 9 \mod 17 = 9$$

Step 3: Compute y_3

$$y_3 = 4(2 - 9) - 7 = 4(-7) - 7 = -28 - 7 = -35 \mod 17$$

$$-35 \mod 17 = 17 \cdot 3 = 51, 51 - 35 = 16 \Rightarrow -35 \equiv 16 \mod 17$$



Final result:

$$(2, 7) + (5, 2) = (9, 16)$$





Elliptic Curve Cryptography (ECC)

2. $(3, 6) + (3, 6)$ — Point doubling

Use doubling formula:

$$\lambda = \frac{3x_1^2 + a}{2y_1} \mod 17$$

Step 1: Compute slope λ

$$x_1 = 3, y_1 = 6$$

$$\lambda = \frac{3(3)^2 + 2}{2 \cdot 6} = \frac{27 + 2}{12} = \frac{29}{12} \mod 17$$

$$29 \mod 17 = 12 \quad \text{and} \quad 12^{-1} \mod 17 = 10 \quad (\text{since } 12 \cdot 10 = 120 \equiv 1 \mod 17)$$

$$\lambda = 12 \cdot 10 = 120 \mod 17 = 1$$

Step 2: Compute x_3

$$x_3 = \lambda^2 - 2x_1 = 1 - 6 = -5 \mod 17 = 12$$





Elliptic Curve Cryptography (ECC)

Step 3: Compute y_3

$$y_3 = \lambda(x_1 - x_3) - y_1 = 1(3 - 12) - 6 = -9 - 6 = -15 \mod 17 = 2$$



Final result:

$$(3, 6) + (3, 6) = (12, 2)$$



